Daniel Precioso, PhD

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Data Scientist

7+ years taking ML/DL projects from concept to production across healthcare, energy, insurance, and maritime logistics. I combine technical rigor (PhD, 5 Q1 publications) with **client-facing delivery**: presenting to executives, deploying production systems (Flask, CI/CD), and translating complex models into business value. Currently coordinating technical delivery for **Vienna Insurance Group**, where I manage stakeholder communications and deploy ML applications for climate risk analysis.

Technical Skills

Languages	:	Python (expert), R, MATLAB, SQL
ML/DL	:	scikit-learn, TensorFlow, PyTorch, Keras, JAX
GenAI & LLMs	:	OpenAI API, Claude, Gemini, Multi-agent systems
MLOps & DevOps	:	Git (CI/CD, pre-commit hooks), Flask, Docker (basics)
Cloud & Data	:	AWS S3, Pandas, NumPy, Streamlit
Academic Tools	:	LaTeX, Overleaf, Blackboard, WordPress

Experience

Postdoctoral Researcher & Adjunct Professor — Sep 2023 – Present
IE University — *Madrid, Spain*

- Coordinate **client-facing technical delivery** for Vienna Insurance Group: manage stakeholder presentations, deploy **production Flask applications** (PythonAnywhere), and implement **CI/CD workflows**. Built climate risk models (DLNMs, Relative Risk curves) used by VIG for portfolio decisions.
- Implemented Kolmogorov-Arnold Networks (KANs) for weather forecasting, improving prediction accuracy.
- Leverage **GenAI tools** (Claude, OpenAI, Gemini) for development acceleration.
- Built and maintained IE Research Datalab's website, increasing research visibility and collaboration opportunities.
- Delivered technical courses to 100+ students with consistently high satisfaction ratings.

COO — Apr 2023 – Present
Canonical Green — *Madrid, Spain*

- Built and deployed route optimization models for maritime clients, demonstrating **5-40% reductions** in fuel consumption through weather-integrated navigation. Managed data workflows using **AWS S3** for large-scale datasets.
- Delivered technical presentations to investors and stakeholders, translating complex algorithms into business value.
- Awarded a prestigious Torres Quevedo Grant (2023) by the Spanish Ministry of Science to develop R&D activities at Canonical Green (grant declined for strategic reasons).

Data Scientist — Sep 2022 – Apr 2023
Komorebi AI — *Madrid, Spain*

- Cleaned and processed 500K+ data points from raw sensor data, enabling accurate fish population monitoring.
- Designed and deployed gradient-boosted and CNN models that achieved 90%+ accuracy in real-time tuna detection.
- Built an interactive dashboard that leadership used daily for operational decisions, reducing analysis time by 80%.
- Implemented **Git workflows with CI/CD practices** for team collaboration on production codebase.

Predoctoral Research Staff Sep 2019 – Aug 2022
University of Cádiz *Cádiz, Spain*

- Built ML models for Industry 4.0 applications in energy monitoring and non-destructive testing, reducing operational costs for industrial partners.
- Partnered with Spain's Health Alerts Centre to create COVID-19 ICU forecasting models that informed national policy decisions during the pandemic.
- Delivered presentations at international conferences, translating complex research into actionable insights for diverse audiences.
- Published 5 peer-reviewed papers in Q1/Q2 journals, advancing the field of applied ML and DS.

Junior Data Scientist Jan 2019 – Jun 2019
Foqum *Madrid, Spain*

- Built neural network models using Keras and TensorFlow for non-intrusive load monitoring (NILM) in energy sector.

Education

Phd in Computer Engineering Sep 2019 – Jul 2023
University of Cádiz *Cádiz, Spain*

MSc in Statistical and Computational Information Processing Sep 2018 – Jul 2019
Universidad Politécnica de Madrid *Madrid, Spain*

Degree in Physics Sep 2014 – Jul 2018
Complutense University of Madrid *Madrid, Spain*

Projects

KAN4Met *Machine Learning, Meteorology* IE University

- Implemented Kolmogorov-Arnold Networks for weather forecasting, advancing ML-based prediction accuracy.
- Contributed to research team that secured funding from Spanish Ministerio de Ciencia, Educación y Universidades.

Insurance and Tech *Data Science, Actuarial Science* IE University

- Built predictive models analyzing climate impact on insurance risk that VIG used for data-driven portfolio strategies.
- Contributed to industry-academic partnership with Vienna Insurance Group (VIG), translating complex climate models into actionable business insights.

Weather Navigation *Mathematical Optimization, Maritime Transport* IE University

- Designed optimization algorithms that reduced maritime fuel consumption by 10% while improving route safety and cutting carbon emissions.
- Contributed to research team that secured substantial funding from BBVA Foundation and Spanish Agencia Estatal (grant TED2021-129455B-I00).
- Built and maintained the official project website https://weathernavigation.com/.
- Presented technology to the European Parliament, positioning the innovation for EU-wide adoption.

Tun-AI *Deep Learning, Fishing Industry* Komorebi AI

- Partnered with Satlink to build AI-powered tuna monitoring systems that automated detection with 90%+ accuracy.
- Transformed 500K+ raw echosounder readings into clean, structured datasets that powered ML models.
- Deployed gradient-boosted trees and CNNs optimized for real-time fish detection on cloud, enabling instant monitoring decisions.

NeoCam *Computer Vision, Healthcare* Univesidad de Cádiz

- Built a contactless monitoring system using Luxonis OAK-D cameras that enabled health experts to track newborn vitals in NICUs.
- Won 2nd place globally at OpenCV AI Competition 2021 among 1,400+ teams.
- Co-inventor on official patent registration in 2023, protecting intellectual property for future commercialization.

COVID-19 *Time Series Forecasting, Healthcare* Univesidad de Cádiz

- Partnered with Spain's Coordination Centre for Health Alerts to build ICU occupancy forecasting models that informed resource allocation during peak pandemic periods.
- Analyzed non-pharmaceutical interventions (NPIs) impact on COVID-19 spread, providing evidence-based recommendations that shaped public health policy.

UCAnFly *Computer Modeling, Astrophysics* Univesidad de Cádiz

- Built MATLAB simulations modeling Earth's electromagnetic field effects on gravitational test mass for space-based detectors like LISA.
- Contributed electromagnetic modeling expertise to multidisciplinary team selected by ESA's Fly Your Satellite! programme.

ATENEA *Computer Vision, Industrial Automation* Univesidad de Cádiz

- Built ML and NLP solutions for Airbus D&S that automated production processes, reducing manual inspection time.
- Contributed to industry partnership funded by CDTI Interconnecta program.

Courses and Certifications

- AI in Education: Leveraging ChatGPT for Teaching (Coursera)
- Machine learning in Python with scikit-learn (France Université Numérique)
- XV Modeling Week at UCM (Coordinator) (UCM)
- Fly your Satellite - 3 CDR Virtual Workshop (ESA)
- TensorFlow in Practice Specialization (Coursera)
- Applied Social Network Analysis in Python (Coursera)
- Applied Machine Learning in Python (Coursera)
- Applied Text Mining in Python (Coursera)
- Introduction to Data Science in Python (Coursera)

Publications

- New analysis in the preliminary design for LNG and LPG ships. Marine Structures, Elsevier, 2025 (Q1).
- HADAD: Hexagonal A-Star with Differential Algorithm Designed for weather routing. Ocean Engineering, 2024 (Q1)
- Hybrid Search method for Zermelo's navigation problem. Computational and Applied Mathematics, 2024 (Q2)
- Aggregation dynamics of tropical tunas around drifting floating objects based on large-scale echo-sounder data. Marine Ecology Progress Series, 2023 (Q1)
- Effectiveness of non-pharmaceutical interventions in nine fields of activity to decrease SARS-CoV-2 transmission (Spain, September 2020-May 2021). Frontiers in Public Health, 2023 (Q1)
- Thresholding Methods in Non-Intrusive Load Monitoring. The Journal of Supercomputing, 2023 (Q2)
- NeoCam: An edge-cloud platform for non-invasive real-time monitoring in neonatal intensive care units. IEEE Journal of Biomedical and Health Informatics, 2023 (Q1)
- TUN-AI: Tuna biomass estimation with Machine Learning models trained on oceanography and echosounder FAD data. Fisheries Research, 2022 (Q2)